Policy Learning Costs and Firm Dynamics

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Using comprehensive Finnish employer–employee register data, I document that furlough use is highly concentrated across firms: most firms never furlough workers, and firm fixed effects explain several times more variation in furlough rates than industry and year effects. Moreover, past furlough use strongly predicts future furlough use. I develop a firm dynamics model that addresses the persistence through learning and use it to discipline a quantitative analysis of furlough policy. The cost structure makes the choice between furloughs and layoffs depend on the persistence of the underlying shock policy know-how in each policy. I calibrate the model to the Finnish data and find that the learning channel explains the empirical patterns. In the counterfactual exercise I show that [ADD]

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1 Introduction

Do firms take up policies that could help them? Can firms process all available information to use policies effectively, and how costly is it to do so? One example of such a policy is furloughing. A firm can choose to use the policy, as opposed to, for example another margin of reducing workforce, layoffs, which is more directly governed by a firm's production function. In comparison to layoffs, furloughs are easily identifiable on the data, as opposed to layoffs, which are harder to detect from voluntary movement of the workers. Furloughs, alongside similar policies such as short-time work, help firms during temporary difficulties by exempting them from payroll costs while preserving the match between firm and worker. The policy should be especially useful for firms that face liquidity constraints, such as small firms. Yet survey evidence from Finland shows that especially small firms find the legislation of the furlough scheme confusing (Alasalmi et al., 2024).

This paper documents a set of empirical patterns about which Finnish firms use the furlough scheme and shows they are consistent with firms facing a fixed cost the first time they furlough. I build a heterogeneous-firms model in the tradition of Hopenhayn and Rogerson (1993), extended with a furlough margin and a fixed learning cost, to discipline these patterns quantitatively. A learning cost — the time, attention, and expertise to understand the legislation and set up internal procedures — is one microfoundation for the fixed cost, supported by survey evidence.

Empirically, year and 3-digit industry fixed effects together absorb only a small share of the variation in the firm-level furlough rate, while firm fixed effects absorb roughly seven times as much, meaning that persistent firm-level heterogeneity is the dominant source of variation compared to the business cycle or industry. Furloughs are highly concentrated: 65% of firms never furlough. The furlough probability at age 5-10, given that a firm has furloughed in its first five years is 42% while it is only 9% if the firm does not have furlough experience. Moreover, volatile firms furlough more, indicating that learning a tool to tackle the uncertainty are valuable to firms. The empirical evidence described in this paragraph control for firm size and productivity. As a robustness check I find that financial constraints are the the main predictor of furlough intensity, and that seasonality does not drive the results.

The model builds on Hopenhayn and Rogerson (1993) with three additions. First, firms have access to a furlough margin in addition to the standard hiring–firing choice: in any period, a firm can keep a worker on the payroll but not engaged in production, not paying wage but a convex per-period reorganization cost. This allows the firm to produce higher amounts in the next period without having to pay hiring costs. Second, using either the furlough margin or the layoff margin freely or at a low cost requires that the firm have prior experience with that margin; the first use of each margin incurs a one-time administrative cost, interpreted as the learning cost. Third, accumulated experience can be lost: in any period a firm leaves a margin idle, it loses experience with some probability, independently of how long ago it was acquired. This stochastic forgetting of policy know-how captures management turnover and

regulatory change.

The main contribution is to document that furlough use is concentrated in ways that are difficult to reconcile in a model without learning, and to formalize a fixed learning cost as one tractable assumption for those patterns. In the calibrated model, [ADD]. The model formalizes firm-side costs and usage of furloughs and links that to the extent to which firms use labor market policies. The Finnish institutional setting cleanly allows to study firm behavior because firms are fully relieved from payroll costs, which isolates them from financial incentives other than those related to firm's own production. For example, in Italy, short-time work is applied on average to low-productivity workers (Giulia Giupponi, 2023), in part because firms pay 4–8% of short-time wages and receive capped compensation (ETUC, 2020), which directs the policy toward low-wage workers, whereas in Austria, furloughed workers are high-productivity ones (Nekoei and Weber, 2020).

A strand of literature focuses on regulation costs to firms. For example, Kalmenovitz (2022) measures the cost of compulsory regulatory compliance and shows that it incentivizes firms to invest less and hire fewer employees, while a related macro literature treats regulation as a tax-like wedge that reduces productivity or entry (Hopenhayn and Rogerson, 1993; Hsieh and Klenow, 2009; Restuccia and Rogerson, 2008; Akcigit and Ates, 2021). This literature has focused on compulsory regulations and their aggregate consequences. I instead study a voluntary policy and show that the cost of learning to use it, rather than the cost of complying with involuntary policies (e.g. following accounting regulation), drives its adoption.

This paper is related to literature that studies temporary layoffs, furloughs and short-time work. Feldstein (1976) sets the theoretical foundations for temporary layoffs based on preserving employer-employee match. Early empirical literature focuses on cyclicality and seasonality (Lilien, 1980; Card and Levine, 1994), while more recent empirical research highlights the role of temporary layoffs in economic recovery and labor market efficiency (Nekoei and Weber, 2020). In the US small businesses were found to be less aware of the Paycheck Protection Program and consequently less likely to use it (Humphries et al., 2020), which is direct evidence that information frictions shape policy take-up. Macro-labor literature focuses on recalls, or loss of recall (for example, Gertler et al. (2026)).

The paper is structured as follows: Section 2 outlines the main features of the Finnish furlough system. Section 3 introduces the dataset and describes the construction of some key variables. Section 4 presents empirical evidence consistent with a fixed cost in furlough use. The model is introduced in Section 5, while the results are in Section 6. Finally, Section 7 concludes. The Appendix contains technical details and extra material.

2 Institutional Background

In Finland furloughed workers have an employment contract with the firm, but do not provide any work and the firm ceases wage payments. The worker usually receives earnings-related unemployment insurance regardless of whether they are furloughed or laid off, with 80% replacement rate for workers with lowest salary and the replacement rate decreases with income[1]. Unemployment insurance and benefits are collectively funded by the government, firms, workers and unions. That is, there is no experience rating.

Before making any decisions about furloughs or layoffs, a firm must start restructuring negotiations that thoroughly assess the need for employment restructuring[2]. Instead of decreasing workforce, the negotiations may result, for example, in reorganizing workforce or in retraining workers from obsolete to productive positions. Firms may offer a different wage if the job positions changes. The negotiations last from 14 days to six weeks depending on whether fewer than ten or more than ten workers are laid off. Firms with fewer than 20 workers are exempt from lengthy negotiations. Furloughs are possible in two cases: i) a firm has a permanent need to lay off workers but decides to furlough instead, or ii) it experiences a sudden and transitory (less than ninety days) decrease in labor demand. However, in case the firm has a sudden and urgent need to furlough, a 14[3] day notice period is sufficient instead of the lengthy negotiations.

A furlough can be fixed-term or open-ended and full-time or part-time, which resembles short-time work schemes in many European countries. In Finland it is more common to be furloughed full time. If the furlough has no agreed end date, the worker may resign without a notice period. Moreover if the furlough has lasted over two hundred days, the worker receives notice period pay without having to provide work for the firm. This gives the firm incentives to recall within the given period. With an open-ended furlough workers can be recalled with a seven-day notice. Furloughed workers can work elsewhere while being furloughed. A firm is obliged to recall workers it has laid off up to six months after the layoff, if the worker's skills match the newly opened vacancy. If a firm has furloughed and laid off, the furloughed workers must be recalled first.

The workers have some incentives to stay furloughed. First, new jobs usually have at least four months of probationary period during which the person can be dismissed without reason. Therefore the worker perceives the old job as more secure even though it might contain furlough periods. Second, the worker does not have to bear search cost and cost of training for a new job. As the furlough contract binds both parties, furlough is more credible than the US style temporary layoff, which might end in loss of recall. If the worker voluntarily quits, they are not eligible for unemployment insurance for a few months. The

[1] Workers who are not eligible collect unemployment benefits, which are often significantly lower

[2] The law was updated in 2021, and this section considers the law before these changes took place.

[3] Some industries have a notice period of one month due to union regulations.

unemployment insurance is the same for furloughed and laid off workers.

3 Data

The paper uses linked employer-employee data to match furloughs to firms. Employment and furlough data are provided by the Ministry of Employment and the Economy of Finland. Firm data are based on financial statements and business registers administered by Statistics Finland.

The employment data contains all employment periods and often has gaps between two consecutive periods with the same employer-employee pair. An employer-employee pair is defined based on the first and last time the pair is observed, which allows for overlapping employment periods. To match furloughs to employment data, the furlough must start within the employer-employee period or within two weeks after the employment has ended. Since this can result in furloughs being linked to multiple employer-employee pairs, another rule is applied that the furlough must start within one day after the employment ends. This ensures that the furlough is not at the beginning of a job obtained after being furloughed. Moreover, the employer-employee pair must not be a side job if it connects to more than one firm. Furloughs that link the worker to more than one firm after this procedure are dropped.

The furlough data, measured on a daily level, is accurate for the start date of the furlough, as furloughed employees must register with employment services to receive benefits. The end date, however, is less accurate and the data might exaggerate the length of furloughs. Before aggregating the furlough data annually for each firm, furloughs lasting over a year are truncated to a maximum duration of one year. On top of this I winsorize at the 99th percentile, annually. Total annual furloughed workforce for each firm represents the number of full-time-equivalent workers on furlough.

The workforce of a firm is measured as the sum of its paid employees and furloughed workers. Since a decrease in workforce is measured as the difference in workforce between two consecutive years, I must include furloughed workers; otherwise, the difference would reflect changes in furloughed workforce rather than actual workforce reductions. Workforce changes of less than one worker are not counted as a decrease in workforce, to exclude small or unintentional variations. Decrease in workforce is defined to be a layoff.

Agriculture, finance, public sector, organizations, households and firms with unknown sectors are excluded due to data issues. Further, public sector, organizations and households are excluded as the focus is on competitive firms[4]. In education industry only competitive firms that do not in general receive subsidies are included. Firms must have at least €10,000 in annual revenue and must report positive assets, wage bills, value added, and social security payments. All monetary variables are deflated to 2021 euros. Each firm must have at least one full time worker.

[4]Limited liability, partnership and limited partnership

4 Empirical Evidence

In this section I present the empirical evidence that motivates the model. Before proceeding, I will provide some background information of furlough use in Finland, presented in Alasalmi et al. (2024). Overall, furloughs stay below 1% of workforce in non-crisis years and spike to nearly 3% in 2020. Averaged across the full sample, only 1.3% of the workforce is on furlough in a typical year. Approximately 10–18% of firms use furloughs during a regular year and in crisis years the fraction increases, up to 32% in 2020. Some industries use furloughs more than others, but on average the fraction on furloughs to workforce is 1-4% in each industry. When comparing furloughs to reduction in labor force within an industry, the figures are between 10–32%, meaning that up to a third of labor force reduction is in furloughs[5]. In construction nearly 40% of reduction in labor force is done using furloughs. When a firm furloughs, it is usually a small fraction - about 17% furlough less than 1 % of workforce. Nearly 20% of established firms furlough more than every other year and 26% furlough at most every ten years. Descriptively, the survival probability over five years for firms that make very large furloughs is similar to firms that do not experience major reductions in workforce.

One important assumption the paper builds on is survey evidence. Alasalmi et al. (2024) reports that the survey was administered to Finnish firms and asked, among other things, about their experience with the furlough scheme. The most striking finding is that small firms report that furloughing is difficult to implement and that the legislation is complicated. Firms with fewer than five workers make up roughly two thirds of the firm-year observations in the register data, so the population of firms that finds the scheme confusing is large.

Empirical Strategy

To study the intensive and extensive margin[6], I estimate the following equation:

$$furlough_{j,t} = \alpha + \beta_1 \, \mathbf{X}_{j,t} + \mathbf{Z}'_{j,t}\boldsymbol{\beta}_2 + \varepsilon_{j,t}, \tag{1}$$

where $furlough_{j,t}$ is the share of firm j's workforce on furlough in year t expressed in percent (so a value of 1 corresponds to 1% of the firm's workforce being on furlough that year), or a dummy indicating whether a firm has furloughed that year. $\mathbf{X}_{j,t}$ contains firm-level controls described in each subsection, and $\mathbf{Z}_{j,t}$ denotes year, 3-digit industry, and firm fixed effects, included incrementally across columns. Standard errors are clustered at the firm level.

[5] This is the upper bound as the length of furloughs is less reliable information compared to the start date of furlough.

[6] The empirical evidence is mostly presented in the intensive margin for more straightforward interpretation. Extensive margin results are in the Appendix.

4.1 Concentration, Fixed Effects and Probability to Furlough

Only a minority, 35%, of firms ever furlough, which suggests that furloughs are not randomly distributed in the economy, but that there is strong selection into furloughs. Another 11–12% of firms furlough around 27% of their years, as evident in Figure 1. Only a small minority of firms furlough in many years, as the thin right tail suggests.

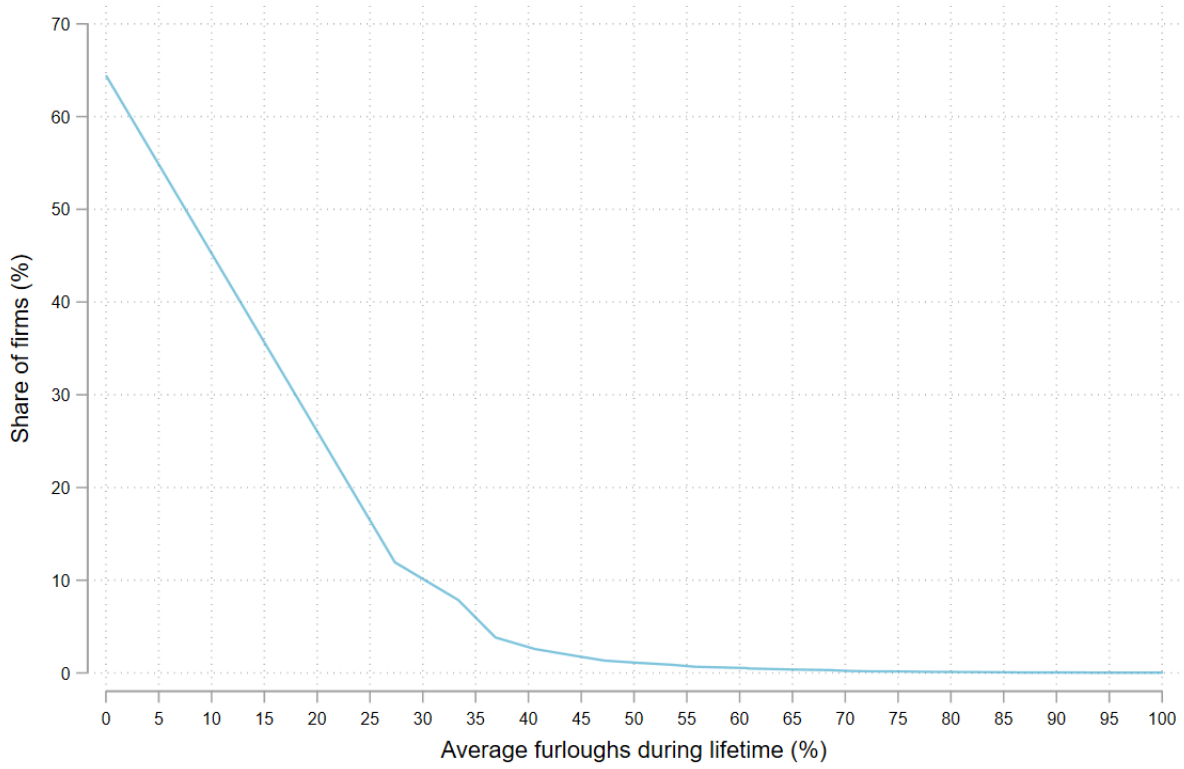


Figure 1: Fraction of firms that furlough over the years 1994-2021. 65% of firms never furlough, while around 11% of firms furlough around a third of their lifetime

The second pattern shows that although business cycle and industry have a role in furloughs, firm fixed effects are the most prominent factor in explaining variation in furlough probability and furlough rate.

Tables 1 and 2 report the R^2 from regressing furlough rate or furlough probability on a constant alone (column 1), on year fixed effects only (column 2), on 3-digit industry fixed effects only (column 3), and on firm fixed effects only (column 4). Each dimension of heterogeneity is included in isolation so that the R^2 can be read as the share of variation in $furlough_{j,t}$ that this dimension alone absorbs. The unconditional mean furlough rate is 1.32% and the furlough probability is 16.5%. Conditional on the firm furloughing at all in year t, the mean furlough rate is 8.04% (Table 12 in the Appendix).

Table 1: Furlough robability

	(1)	(2)	(3)	(4)
Furlough probability	0.165***	0.165***	0.166***	0.166***
	(0.0039)	(0.0037)	(0.0020)	(0.0018)
Observations	1,885,924	1,885,924	1,885,924	1,850,793
R-squared	0.000	0.015	0.097	0.428
Year FE		✓		
Industry FE			✓	
Firm FE				✓

Results from a linear probability model. Industry at 3-digit level. Standard errors are clustered at firm level. *p<0.10, **p<0.05, ***p<0.01.

Table 2: Furlough rate

	(1)	(2)	(3)	(4)
Furlough rate	1.324***	1.324***	1.324***	1.331***
	(0.0450)	(0.0410)	(0.0300)	(0.0276)
Observations	1,885,924	1,885,924	1,885,924	1,850,793
R-squared	0.000	0.015	0.049	0.355
Year FE		✓		
Industry FE			✓	
Firm FE				✓

Industry at 3-digit level. Standard errors are clustered at firm level. *p<0.10, **p<0.05, ***p<0.01.

Recurring Furloughs. If a firm ever furloughs, its first furlough tends to be in its early years[7]. The high concentration of first furlough in the early years confirms that selection into furlough scheme is strong and that firms learn in their early years. Tables 3 and 4 show that firms that have furloughed in the early years tend to do so in the later years, much more than firms that do not furlough in the early years. The difference is substantial. Controlling for industry and year, firms that have the experience furlough 42% of the time while firms that do not have the experience furlough only around 9% of the time.

These patterns, however, could be a product of other persistent firm-level drivers, such as business model, or worker composition, and the remaining subsections work to narrow down which of these are likely to be at play.

[7]See Figure 4 in the Appendix. Roughly 9% of eventual furloughers furlough for the first time in their first operational year, and the hazard declines monotonically with firm age, falling to around 2% by firm age 20 and remaining roughly flat thereafter.

Table 3: Furlough probability given that the firm has not furloughed in its early years

Furlough probability	(1)	(2)	(3)
Constant	0.088***	0.088***	0.088***
	(0.0006)	(0.0006)	(0.0006)
Observations	248,493	248,493	248,492
R^2	0.000	0.020	0.054
Year FE		✓	
Industry FE			✓

Dependent variable is the furlough probability over the next five years, given that the firm has not furloughed in its first five years of life. Early years refers to the first five years of the firm's lifetime. Industry at 3-digit level. Standard errors are clustered at firm level. *p<0.10, **p<0.05, ***p<0.01.

Table 4: Furlough probability given that the firm has furloughed in its early years

Furlough probability	(1)	(2)	(3)
Constant	0.422***	0.422***	0.422***
	(0.0020)	(0.0020)	(0.0019)
Observations	62,678	62,678	62,678
R^2	0.000	0.013	0.090
Year FE		✓	
Industry FE			✓

Dependent variable is the furlough probability over the next five years, given that the firm has furloughed in its first five years of life. Early years refers to the first five years of the firm's lifetime. Industry at 3-digit level. Standard errors are clustered at firm level. *p<0.10, **p<0.05, ***p<0.01.

4.2 Larger and Less Productive Firms Furlough More

While larger and less productive firms furlough more, the effect is moderate. Table 5 regresses the furlough rate on log employees and log labor productivity. A 10% increase in firm size is associated with a 0.010 to 0.017 percentage-point increase in the furlough share, depending on the fixed effects. Relative to the sample mean of 1.32, this is about 0.76-1.3%. The productivity coefficient is negative and larger in absolute terms: a 10% increase in labor productivity decreases furlough intensity by 0.046-0.063 p.p., which is 3.5-4.8% of the mean.

Importantly, the magnitudes survive the inclusion of firm fixed effects (column 4), and in fact grow somewhat in size. The size coefficient rises from 0.133 to 0.168 and the productivity coefficient from

−0.514 to −0.631. The relationships therefore reflect within-firm variation, not just sorting of firms across persistent size and productivity bins. Firms become more likely to furlough in years when they are larger or less productive than usual.

Table 5: Furlough rate by size and productivity

Furlough rate	(1)	(2)	(3)	(4)
Employees	0.133***	0.131***	0.100***	0.168***
	(0.0093)	(0.0096)	(0.0113)	(0.0225)
Labor productivity	-0.514***	-0.456***	-0.545***	-0.631***
	(0.0376)	(0.0283)	(0.0281)	(0.0366)
Constant	1.267***	1.255***	1.324***	1.253***
	(0.0450)	(0.0306)	(0.0285)	(0.0419)
Observations	1,885,924	1,885,924	1,885,924	1,850,793
R^2	0.004	0.019	0.053	0.358
Year FE		✓		
Industry FE			✓	
Firm FE				✓

Industry at 3-digit level. Standard errors are clustered at firm level. *p<0.10, **p<0.05, ***p<0.01.

4.3 Financial Frictions

A natural channel alternative to learning is that firms differ in their access to short-term finance. For example, Giupponi et al. (2022) show that liquidity-constrained firms are more likely to take up short-time-work schemes during downturns. If furloughs are mechanically a substitute for borrowing through bad years, then the size and productivity gradients documented above might simply reflect financial constraints rather than information frictions. This subsection tests that alternative by adding standard balance-sheet measures of liquidity and leverage to the firm-fixed-effects specification of Table 5.

Table 6 adds, one at a time, six financial variables: a liquidity, the cashflow-to-assets ratio, the equity ratio (%), log short-term liabilities, log inventories, and the firm's implied interest rate[8]. Two patterns stand out.

First, the size and productivity coefficients are essentially unchanged across columns, with one informative exception. The size coefficient is approximately 0.168 across columns 1–3, and the productivity coefficient is also stable in those columns. The exception is column 4, where adding log short-term liabilities raises the size coefficient to 0.309. Having more short-term liabilities decreases furlough intensity. A very high

[8]Calculated from the data as interest payments divided by debt

one-period loan could indicate that the firm is heavily investing, and thus production capability is in full use.

Second, the financial variables themselves are mostly weak predictors. The liquidity, cashflow-to-assets, and equity ratio coefficients are zero. The interest rate coefficient (column 6) is similarly tiny in economic terms: with the rate in decimals, even a five-percentage-point ($\Delta = 0.05$) shift in the firm's borrowing rate would move the furlough share by less than 10^{-3} percentage points within firm. Two coefficients are larger in value: log short-term liabilities (column 4, -0.166) and log inventories (column 5, -0.024). For short-term liabilities, a 10% increase is associated with a 0.017 percentage-point lower furlough share; for inventories, the corresponding magnitude is 0.002 percentage points. These variables may provide an alternative to having to furlough by meeting short-term needs by borrowing or running down inventory.

Table 6: Furlough rate with various liquidity and leverage measures

	Furlough rate					
	(1)	(2)	(3)	(4)	(5)	(6)
Liquidity	-0.000					
	(0.0000)					
Cashflow		0.000*				
		(0.0001)				
Equity ratio			-0.000			
			(0.0000)			
Short-term liabilities				-0.166***		
				(0.0130)		
Inventories					-0.024**	
					(0.0108)	
Interest rate						0.001**
						(0.0002)
Size	0.167***	0.168***	0.168***	0.309***	0.103**	0.161***
	(0.0226)	(0.0225)	(0.0225)	(0.0300)	(0.0390)	(0.0422)
Productivity	-0.633***	-0.631***	-0.631***	-0.636***	-0.729***	-0.839***
	(0.0366)	(0.0366)	(0.0366)	(0.0376)	(0.0519)	(0.0591)
Constant	1.255***	1.253***	1.253***	2.937***	1.689***	1.407***
	(0.0421)	(0.0419)	(0.0419)	(0.1265)	(0.0951)	(0.0714)
Observations	1,843,766	1,850,776	1,850,793	1,794,551	1,081,414	1,115,162
R^2	0.358	0.358	0.358	0.362	0.363	0.394

Regression includes firm fixed effects. Standard errors are clustered at firm level. *p<0.10, **p<0.05, ***p<0.01.

4.4 Revenue Shocks and Volatility Predict Furloughs

Volatility raises incentives for firms to learn to furlough, in the spirit of rational-inattention models in which firms acquire information about policies they expect to use repeatedly (Mikosch et al. 2024; Paciello and Wiederholt 2014). The more volatile revenue a firm experiences, in short or long run, the more intensely the firm furloughs.

Persistent revenue volatility. Firms whose revenue is persistently volatile use furloughs more. Table 7 replaces the contemporaneous shock with a 3-year rolling standard deviation of change in log revenue at firm-level. The coefficient is positive throughout but decreases with firm fixed effects: with firm FE (column 4), a 10% increase in the rolling volatility is associated with a 0.07-0.2 percentage-point increase in the furlough share, about 15% of the unconditional mean. The within-firm coefficient is smaller than the population-level one (0.71 vs. 1.97), so a substantial part of the volatility-furlough relationship reflects sorting. Volatile firms are different from non-volatile firms in ways that fixed effects absorb.

Table 7: Furlough rate and persistent revenue volatility

Furlough rate	(1)	(2)	(3)	(4)
SD of Δ log revenue	1.970***	1.991***	1.249***	0.712***
	(0.0874)	(0.0817)	(0.0603)	(0.0535)
Size	0.074***	0.062***	0.020***	-0.030
	(0.0083)	(0.0080)	(0.0094)	(0.0270)
Productivity	-0.694***	-0.590***	-0.742***	-0.879***
	(0.0493)	(0.0372)	(0.0375)	(0.0502)
Constant	1.009***	0.997***	1.281***	1.524***
	(0.0424)	(0.0378)	(0.0256)	(0.0530)
Observations	1,368,719	1,368,719	1,368,719	1,349,028
R^2	0.018	0.036	0.066	0.393
Year FE		✓		
Industry FE			✓	
Firm FE				✓

Volatility measured as 3-year rolling standard deviation of Δ log revenue. Industry at 3-digit level. Standard errors are clustered at firm level. *p<0.10, **p<0.05, ***p<0.01.

4.5 Furloughing as a Business Plan

A different concern is that the firm-level concentration documented above is mechanically driven by a small group of firms for which furloughing is part of the regular business model, like seasonal sectors that furlough every winter, for example. Such firms would generate strong firm fixed effects without paying

any learning cost in the sense the model emphasizes because they simply renew the same labor-adjustment routine year after year. To check this, I re-estimate all specifications above on a sample that excludes firms that furlough in more than 80% of the years they are observed. The results are unchanged, which indicates that the patterns are not driven by habitual furloughers. Refer to the Appendix for the regression estimates.

4.6 Discussion

Taken together, the facts are hard to reconcile in a model that does not account for persistence in furlough behavior. If furloughing were a low-cost margin uniformly available to all firms, furlough activity would be spread evenly across the firm population, and past use would not predict future use so strongly. These empirical observations point to a cost that makes firms select into furloughs.

The remaining facts further explore the relationship between firm characteristics and furloughs, to ensure that the model considers all relevant aspects of the data. The size and productivity gradients are modest. This The within-firm volatility result reinforces this interpretation. A firm that has just gone through a volatile stretch has more reason to invest in understanding the scheme than one that has had a stable run, and because this is a within-firm result, it cannot be attributed to fixed firm characteristics such as industry or management style. That the concentration survives excluding habitual furloughers rules out the alternative that these patterns simply reflect a business model rather than a cost paid once. The robustness analysis about financial channels shows that although some financial variables are weakly correlated with furlough rate,

5 Model

5.1 Environment

The model is built in the tradition of Hopenhayn and Rogerson (1993), featuring a continuum of heterogeneous firms in discrete and infinite time that enter and exit endogenously, face idiosyncratic productivity shocks, and make dynamic employment decisions subject to adjustment frictions. Time period is set to one year. The framework introduces two key departures from the standard model. First, firms have access to furloughs as an intensive-margin labor adjustment tool. Rather than laying off workers when hit by a negative idiosyncratic productivity shock, a firm can temporarily reduce the fraction of its workforce engaged in production while keeping workers on the payroll. Second, firms accumulate extensive margin experience in using furloughs and layoffs, which reduces the administrative costs of these actions in future periods. Experience is, however, subject to stochastic forgetting: in any period in which a margin is left idle, the firm loses its experience with some fixed probability, independently of how long ago it was acquired. This captures the idea that organizational knowledge about how to conduct furloughs or layoffs can be lost when the practice is not regularly used, for instance due to management turnover, or due to changes in regulation.

The model builds on Hopenhayn and Rogerson (1993) with three additions, mapping the model to the empirical patterns. First, firms have access to a furlough margin in addition to the standard hiring–firing choice. In any period, a firm can keep a worker on the payroll but not engaged in production, paying no wage but bearing a convex per-period reorganization cost. Second, using either the furlough margin or the layoff margin freely requires that the firm have prior experience with that margin; the *first* use of each margin incurs a one-time administrative cost. This modeling choice could be interpreted as a learning cost. A convex furlough cost per worker circumscribes the firm from using furloughs instead of layoffs in the case of a permanent negative shock, and having a convex hiring and firing cost prevents firms from always using layoffs. Third, accumulated experience can be lost: in any period a firm leaves a margin idle, it loses experience with some probability, independently of how long ago it was acquired. This stochastic forgetting captures management turnover and regulatory change, and prevents the steady-state share of experienced firms from going to one.

5.1.1 Productivity Shock

Each firm draws an idiosyncratic productivity shock s_t that evolves according to an AR(1) process:

$$s_t = \rho s_{t-1} + \varepsilon_t, \qquad \varepsilon_t \sim N(0, \sigma_\varepsilon^2), \tag{2}$$

where $\rho \in (0, 1)$ governs the persistence of the shock and ε_t is an i.i.d. innovation. Shock persistence ρ plays a central role in the firm's choice between furloughs and layoffs. When a firm is hit by a strong

negative shock, the optimal response depends on how long the shock is expected to last. If the shock is highly persistent, the firm expects to remain unproductive for many periods and may prefer to lay off workers permanently rather than pay furlough adjustment costs each period. Conversely, if the shock is expected to be temporary, the firm prefers to furlough, retaining workers who will again be productive once the shock reverts, and avoiding the future hiring and adjustment costs that a layoff would entail.

5.1.2 Attrition

A fraction $1 - \lambda$ of the workforce leaves the firm each period exogenously, through quits or retirements, independent of the firm's decisions. The parameter $\lambda \in (0, 1)$ is therefore the survival fraction of the workforce. The state variable n is the workforce chosen last period that enters the current period, before attrition occurs. At the start of each period, attrition reduces this to λn, which becomes the baseline from which the firm makes its new employment choice n'. A firm that neither hires nor fires beyond attrition will therefore carry workforce λn into production.

5.1.3 State Space

The payoff-relevant state of an incumbent firm is (s, n, z_f, z_l), where s is the current productivity shock and n is the workforce carried into the period from last period's choice, before attrition occurs. $z_f \in \{0, 1\}$ ($z_l \in \{0, 1\}$) indicates whether the firm has prior experience with furloughs (layoffs) and remembers it. The firm chooses next-period employment $n' \geq 0$ from the post-attrition baseline λn, and the fraction of the workforce engaged in production, i.e. not furloughed, $\gamma \in (0, 1]$.

5.2 Technology and Costs

5.2.1 Production

The firm produces output according to:

$$f(s, n', \gamma) = s(\gamma n')^{\alpha}, \qquad \alpha \in (0, 1), \tag{3}$$

where $\gamma n'$ is the number of workers actively engaged in production. When $\gamma = 1$ there are no furloughs and all hired workers produce. When $\gamma < 1$, a fraction $1 - \gamma$ of the workforce is on furlough and does not contribute to output. The firm sells output at price p and faces a normalized wage $w = 1$.

5.3 Labor Adjustment and Furlough Costs

The combined cost function for an incumbent firm is:

$$c(n', \gamma; n) = \frac{\xi_n}{2} \frac{(n' - \lambda n)^2}{(n' + \lambda n)/2} + \frac{\xi_\gamma}{2} (1 - \gamma)^2 n' \tag{4}$$

For an entrant which begins with no existing workforce, there is no meaningful baseline from which to measure a workforce deviation. The quadratic adjustment cost scaled by λn is therefore undefined for entrants. Instead, entrants face a simpler cost function that retains the wage bill and furlough cost components but replaces the size-scaled hiring cost with an unscaled quadratic cost of building a workforce from scratch:

$$c_0(n', \gamma) = \xi_n n' + \frac{\xi_\gamma}{2} (1 - \gamma)^2 n' \tag{5}$$

The components of $c(\cdot)$ capture distinct economic forces. The first term is a quadratic adjustment cost for changing the size of the workforce relative to the attrition-adjusted baseline λn. Scaling by λn ensures that the friction is proportional to firm size, so that a large and a small firm face the same per-worker cost when adjusting by the same fraction of their workforce. The second term is a convex cost of furloughing, increasing in the share of workers on furlough $(1 - \gamma)$ and scaled by n'. This is the only cost a firm faces when furloughing workers, on top of the fixed administrative cost, described below. The convex terms capture the idea that as a higher share of the workforce is furloughed or laid off, the firm must reorganize tasks and reallocate work among a shrinking active workforce, and these reorganization costs rise with the furlough/layoff share.

5.3.1 Administrative Costs

In addition to the flow costs above, firms incur one-time administrative costs the first time they use a given adjustment margin. Let $\tau_f > 0$ denote the administrative cost of furloughing for the first time, and $\tau_l > 0$ the administrative cost of laying off workers for the first time. A layoff is defined as reducing the workforce below the attrition-adjusted level, i.e. $n' < \lambda n$. These costs capture setup costs such as legal fees, HR procedures, and the organizational learning required to implement a new type of labor adjustment. Once a firm has incurred these costs and has experience ($z_f = 1$ or $z_l = 1$), it does not pay them again, unless it subsequently forgets — the stochastic forgetting process is described in Section 5.4.

Formally:

$$g(z_f, z_l, \gamma, n', n) = \begin{cases} \tau_f + \tau_l & \text{if } \gamma < 1,\ z_f = 0,\ n' < \lambda n \text{ and } z_l = 0 \\ \tau_f & \text{if } \gamma < 1,\ z_f = 0, \text{ and } (n' \geq \lambda n \text{ or } z_l = 1) \\ \tau_l & \text{if } (\gamma = 1 \text{ or } z_f = 1),\ n' < \lambda n,\ z_l = 0 \\ 0 & \text{if } (\gamma = 1 \text{ or } z_f = 1) \text{ and } (n' \geq \lambda n \text{ or } z_l = 1) \end{cases} \tag{6}$$

The first case arises when a firm simultaneously furloughs and lays off workers for the first time, incurring both administrative costs. The second case covers a firm that furloughs for the first time but does not lay off (or already has layoff experience), paying only τ_f. The third case covers a firm that lays off for the first time but either does not furlough or already has furlough experience, paying only τ_l. The final case covers all situations where the firm either does not use a margin or has prior experience with every margin it uses, incurring no administrative costs.

The model treats the furlough and layoff margins symmetrically: each carries its own one-time administrative cost (τ_f, τ_l) and its own forgetting probability (δ_f, δ_l, defined below). This symmetry is structural rather than empirical. The reduced-form patterns documented in Section 4 discipline the furlough parameters τ_f and δ_f directly, while the layoff parameters τ_l and δ_l are not the focus of the empirical exercise and are not separately identified by the moments used here. In the baseline calibration I therefore set the layoff parameters to values that make the layoff-learning channel quantitatively negligible, so that the model speaks unambiguously to the furlough story while keeping the symmetric structure for completeness and possible extensions.

5.4 Experience Dynamics

5.4.1 Accumulation and Forgetting

Each period a firm either uses a labor adjustment margin or leaves it idle. A firm that uses a margin this period immediately acquires experience in it. A firm that leaves a margin idle faces a Bernoulli forgetting shock each period: with probability δ_j it loses its experience, and with probability $1 - \delta_j$ it retains it, where $j \in \{f, l\}$. The parameters δ_f and δ_l can be calibrated separately, allowing the model to match independent empirical moments on furlough and layoff dynamics. This specification captures the idea that know-how in conducting furloughs or layoffs can be lost when the practice is not regularly used, for instance due to management turnover or regulatory changes.

The law of motion for furlough experience is:

$$\Pr(z_f' = 1 | z_f, \gamma) = \begin{cases} 1 & \text{if } \gamma < 1 \\ 1 - \delta_f & \text{if } z_f = 1 \text{ and } \gamma = 1 \\ 0 & \text{if } z_f = 0 \text{ and } \gamma = 1 \end{cases} \tag{7}$$

where case (i) the firm furloughs this period ($\gamma < 1$), so it acquires or retains experience with certainty; case (ii) the firm does not furlough ($\gamma = 1$) but enters the period with experience ($z_f = 1$), so it retains experience with probability $1 - \delta_f$ and forgets with probability δ_f; case (iii) the firm does not furlough and has no experience ($z_f = 0$), so it remains inexperienced with certainty.

The law of motion for layoff experience is:

$$\Pr(z_l' = 1 | z_l, \gamma) = \begin{cases} 1 & \text{if } n' < \lambda n \\ 1 - \delta_l & \text{if } z_l = 1 \text{ and } n' \geq \lambda n \\ 0 & \text{if } z_l = 0 \text{ and } n' \geq \lambda n \end{cases} \tag{8}$$

where case (i) the firm lays off workers this period ($n' < \lambda n$), so it acquires or retains experience with certainty; case (ii) the firm does not lay off ($n' \geq \lambda n$) but enters the period with experience ($z_l = 1$), so it retains experience with probability $1 - \delta_l$ and forgets with probability δ_l; case (iii) the firm does not lay off and has no experience ($z_l = 0$), so it remains inexperienced with certainty.

5.5 Firm Value and Optimization

5.5.1 Exit

Each period, after choosing (n', γ), a firm decides whether to exit. The timing convention follows Hopenhayn and Rogerson (1993): after production and the realization of the current shock, the firm decides at the end of the period whether to remain or to exit. When exit decision is made at the end of a period, next period profit is discounted, but if the decision is made in the current period, profit is not discounted.

5.5.2 Bellman Equation for Incumbents

An incumbent firm with state (s, n, z_f, z_l) solves:

$$\begin{aligned} W(s, n, z_f, z_l; p) = \max_{\gamma \in (0,1],\ n' \geq 0} \Bigg\{ & pf(s, n', \gamma) - c(n', n, \gamma) - pc_f - \gamma n' - g(z_f, z_l, \gamma, n', n) \\ & + \beta \max\Big(\mathbb{E}_{s'|s} \Big[\mathbb{E}_{z_f', z_l'} \big[W(s', n', z_f', z_l'; p) \big] \Big),\ 0 \Big) \Bigg\}, \end{aligned} \tag{9}$$

where c_f is the per-period operating cost and $\beta \in (0, 1)$ is the discount factor. The flow wage bill, $-\gamma n'$, is for active workers. This period workforce, n' chosen by the firm given the post-attrition baseline λn. The outer expectation integrates over next period's productivity shock s' conditional on s. The inner expectation integrates over the stochastic transitions of the experience states (z_f', z_l'), which depend on whether δ_f or δ_l shocks occur. Following the Hopenhayn–Rogerson convention, the firm chooses (n', γ) and produces in the current period, and at the end of the period chooses between continuation and exit. Both options pay off next period and are therefore discounted by β via the outer max. The firm exits if continuation values is less than zero.

Given (s, n, z_f, z_l), the firm weighs current profits net of adjustment and administrative costs against the continuation value, which reflects both the option value of future furlough and layoff experience and the risk of forgetting accumulated experience.

5.5.3 Bellman Equation for Entrants

Entrants pay a setup cost c_e, begin with no workforce ($n = 0$) and no experience ($z_f = z_l = 0$). They face a cost function defined earlier, c_0. Their value function is:

$$W(s, n, z_f, z_l; p) = \max_{\gamma \in (0,1],\ n' \geq 0} \Bigg\{ pf(s, n', \gamma) - c_0(n', \gamma) - pc_e - \gamma n' - g(0, 0, \gamma, n', 0) + \beta \max\Big(\mathbb{E}_{s'|s} \Big[\mathbb{E}_{z'_f, z'_l} \big[W(s', n', z'_f, z'_l; p) \big] \Big),\ 0 \Big) \Bigg\}, \tag{10}$$

5.6 General Equilibrium

Because firms are price-takers, closing the model requires only a representative household and free entry: neither adds a state variable to the firm's problem. Free entry pins down p; the household's labor-supply condition pins down the wage.

Household. A unit mass of identical households supplies labor and consumes the firm sector's output. Following the indivisible-labor preferences of Rogerson (1988), the household's period problem is

$$\max_{C, L} \ \ln C - aL \qquad \text{s.t.} \qquad pC = wL + \Pi, \tag{11}$$

where L is labor supplied to active production and Π is dividend income from the household's ownership of all firms. Furloughed workers receive a government-administered benefit at replacement rate b, financed lump-sum on the same household; since the household is both the recipient and the taxpayer, the benefit and its financing cancel in (11) and can be omitted without loss.[9] Work disutility falls only on active labor L, since furloughed workers are idle. The static first order condition is

$$\frac{w}{p} = aC. \tag{12}$$

Stationary equilibrium. Let μ be the invariant measure of incumbents over (s, n, z_f, z_l), M the mass of entrants, $G(s)$ the entrant productivity distribution, and normalize $w = 1$. A stationary equilibrium is a price p, quantities (C, L), firm policies $\{n'(\cdot), \gamma(\cdot)\}$, entry mass M, and measure μ such that:

(i) policies solve the incumbent and entrant Bellman equations given p;

[9] Explicitly, $pC = wL + bwF + \Pi - T$ with $T = bwF$ collapses to (11). Because furlough pay never enters the firm's problem either, b does not affect firm policy or the equilibrium allocation.

(ii) free entry holds, $\int W(s,0,0,0;p)\,dG(s)=0$, pinning down p;

(iii) the household satisfies (12);

(iv) the labor market clears, $L=\int \gamma\, n'\, d\mu$;

(v) the goods market clears,

$$C=\int s(\gamma n')^{\alpha}\,d\mu-\int c_f\,d\mu-Mc_e-\int\Big[\tfrac{\xi_n}{2}\tfrac{(n'-\lambda n)^2}{\lambda n}+\tfrac{\xi_\gamma}{2}(1-\gamma)^2 n'+g(z_f,z_l,\gamma,n',n)\Big]d\mu; \quad (13)$$

(vi) μ is invariant under the firm policies and entry at mass M.

Wages and the furlough benefit are transfers within the household and drop out of (13).

5.7 Welfare

Lifetime utility of the representative household, evaluated at a stationary equilibrium, is

$$\mathcal{W}=\sum_{t=0}^{\infty}\beta^t(\ln C-AL)=\frac{\ln C-AL}{1-\beta}. \quad (14)$$

Consumption-equivalent welfare. Let subscript 0 denote the baseline and 1 a counterfactual (e.g. a lower furlough learning cost). The consumption-equivalent welfare gain Δ solves $\ln[(1+\Delta)C_0]-aL_0=\ln C_1-aL_1$, so

$$\Delta=\frac{C_1}{C_0}\exp[-A(L_1-L_0)]-1=\frac{p_0}{p_1}\exp[-A(L_1-L_0)]-1, \quad (15)$$

using $C=1/(ap)$ for the second equality. A comparison of the equilibria decomposes into consumption and labor-disutility channels:

$$\Delta\approx\underbrace{-(\ln p_1-\ln p_0)}_{\text{consumption}}-\underbrace{A(L_1-L_0)}_{\text{labor disutility}}. \quad (16)$$

Furloughed workers, unlike active workers, receive utility from not working.

6 Results

I solve for the stationary general equilibrium of the model and calibrate it to Finnish employer-employee data, targeting moments related to furloughs. Some parameters are taken from the data or the literature, and the remaining ones are estimated by the simulated method of moments. I then use the calibrated economy to quantify the cost of information frictions.

6.1 Calibration

Externally set parameters. Table 8 lists the externally set parameters. The discount factor is 0.94, which corresponds to an annual real interest rate of about 6%, and the labor share is 0.64. Both are standard values in the literature. The wage is the numeraire. The workforce survival fraction, λ, is the share of workers who stay with the firm from one year to the next. It is based on an annual job-to-job transition rate of 5% for workers aged 15 to 74 in Finland (Eurostat[10]) and an annual retirement rate of 2%.[11] The layoff forgetting probability, δ_l, is set to 0.10, meaning that learning layoffs is quite persistent – the probability of forgetting rate 10% when the margin is not used. Layoffs are the main margin through which firms reduce employment when a persistent fall in productivity or demand requires a larger cut than natural attrition allows. Most firms use this margin at some point (42% in the data). Furloughs, in contrast, are an optional and temporary margin. I therefore set δ_l low, so that firms rarely forget how to lay off and rarely pay the layoff setup cost again. Since the layoff setup cost is estimated, the choice of δ_l is reflected in its estimated value. The parameters of the AR(1) productivity process are taken from Hopenhayn and Rogerson (1993). The disutility of work, A, is also set as in Hopenhayn and Rogerson (1993). It pins down the mass of entrants.

Table 8: Fixed parameters

Parameter	Description	Value
β	Discount factor	0.94
α	Labor share	0.64
w	Wage (numeraire)	1
c_f	Fixed operating cost	10
λ	Workforce survival fraction	0.93
δ_l	Layoff-experience forgetting probability	0.10
ρ	AR(1) persistence of log productivity (Hopenhayn–Rogerson 1993)	0.93
$\sigma_{\log z}$	Unconditional SD of log productivity, $= \sqrt{0.53}$ (Hopenhayn–Rogerson 1993)	0.7280
σ_ε	AR(1) innovation SD, $= \sigma_{\log z}\sqrt{1-\rho^2}$	0.2676
a	AR(1) intercept / scale normalization (mean log s; absorbed by GE price)	0.078
A	Disutility of work	0.60

Internally calibrated parameters. Table 9 lists the internally calibrated parameters. I estimate them by the simulated method of moments. The main targets relate to furloughs: the share of firms furloughing

[10] https://ec.europa.eu/eurostat/databrowser/view/lfsi_long_e07__custom_22689882/default/table

[11] Own calculations using data from the Finnish Centre for Pensions and Statistics Finland. https://www.etk.fi/tutkimus-tilastot-ja-ennusteet/tilastot/elakkeensaajat-ja-elakemenot/tilasto-suomen-elakkeensaajista/ and https://stat.fi/fi/julkaisu/cm0xpdzse0fy507w3e30r1hlc

in a given year (17%), the share of firms that ever furlough (36%), furlough intensity conditional on furloughing (8%), and the probability of furloughing in years 6 to 10 conditional on having furloughed or not in the first five years of the firm's life. As shown in Section 4, the difference between these two probabilities is large, 33 percentage points. The entry cost is solved from the free entry condition with the price fixed at $p = 1$. In the counterfactuals, I hold the entry cost at this value and let the price adjust.[12]

Model fit. Table 10 reports the model fit. The model matches the share of firms furloughing in a given year and the share that ever furlough well. Among the untargeted moments, the model matches furlough intensity across all firm-years. The exit rate is about half of its empirical value, and the share of firms that ever lay off is lower than in the data.

Table 9: Internally calibrated parameters

Parameter	Symbol	Value	Role
Furlough setup cost	τ_f	0.30	One-time admin cost, first furlough
Furlough intensity cost	ξ_γ	6.00	Convex cost of furloughing
Layoff setup cost	τ_l	7.50	One-time admin cost, first layoff
Employment adj. cost	ξ_n	0.70	Size-scaled quadratic
Furlough forgetting rate	δ_f	0.30	Prob. of losing furlough experience when idle
Entry cost[a]	c_e	74.50	Solved from free entry, $p = 1$

[a] Not estimated by SMM: solved from the free-entry condition $V_e = 0$ at $p = 1$, given the other parameters.

6.2 Importance of the Learning Channel

Calibrating the model without learning, with fixed costs. As an additional exercise, I remove learning from the model but keep the fixed costs, which firms now pay in every year they furlough or lay off. I recalibrate the model to two targets, the share of firms furloughing in a given year and furlough intensity conditional on furloughing. Table 11 shows that the model matches both targets. The untargeted moments, however, show that the model does not generate persistence. Most firms (81%) furlough at some point, and the probability of furloughing in years 6 to 10 is similar for firms that furloughed in the first five years and firms that did not (15 versus 13%). The remaining gap most likely reflects the persistence of productivity shocks.

[12] This reduces the computational burden of the calibration considerably. With p fixed, the entry cost follows directly from the free entry condition, whereas solving for the price requires a bisection.

Table 10: Moment fit in the baseline model

Moment	Data	Baseline
Targeted		
Share of firm-years furloughing	0.17	0.18
Share ever furloughing	0.36	0.33
Furlough intensity if furloughs	0.08	0.04
P(furlough | early furlough)	0.42	0.64
P(furlough | no early furlough)	0.09	0.05
Untargeted		
Mean employment	15.54	57.81
Exit rate	0.09	0.04
SD of employment	142.90	201.90
Furlough intensity, all firm-years	0.01	0.01
Share ever laid off	0.71	0.57

Model moments are computed from a simulated panel of 1,000 firms over 20,000 periods (1,000 burn-in), comparable to the firm-level panel data. Solved $c_e = 74.50$ at $p = 1$.

7 Discussion

The ability of firms to take up government policies is crucial for the policy maker. Why set up a policy that is not used, or used by few? Evidence from the US, without official furlough policy, speaks for the necessity to have a credible temporary layoff policy (furloughs). An interesting feature of temporary layoffs in the US is that those who are expecting to be recalled but are not, have longer unemployment spells (e.g. Katz and Meyer (1990), **?**), which follows from low search efforts among those who expect to be recalled (Nekoei and Weber, 2015). The US job finding rates are lower for workers who think they will be recalled (Gertler et al., 2026), but end up not being recalled. When the decision is made about the necessity of the system, the question is, how intensively it should be used, to maximize the welfare of consumers. If the cost of furlough is very low, firms that should exit use it to prolong exit, but if the cost is too high, productive firms might end up exiting during uncertain times.

This paper documents a set of empirical patterns about furlough use among Finnish firms over 1994–2021 and uses them to motivate a heterogeneous-firms model with a furlough option and learning dynamics. The empirical patterns would be difficult to reconcile with a frictionless model: firm fixed effects absorb roughly seven times as much variation in the firm-level furlough rate as 3-digit industry, furlough use is concentrated among a minority of firms with 65% never furloughing, persistently volatile firms furlough

Table 11: Moment fit in when learning channel is shut down

Moment	Data	No learning
Targeted		
Share of firm-years furloughing	0.17	0.16
Furlough intensity if furloughs	0.08	0.08
Untargeted		
Share ever furloughing	0.36	0.81
P(furlough | early furlough)	0.42	0.15
P(furlough | no early furlough)	0.09	0.13
Mean employment	15.54	64.54
Exit rate	0.09	0.02
SD of employment	142.90	263.87
Furlough intensity, all firm-years	0.01	0.01
Share ever laid off	0.71	0.70

The model has no memory: the fixed costs τ_f and τ_l are paid in every year the firm furloughs or lays off. Calibrated: $\tau_f = 0.052$, $\xi_\gamma = 4.09$. Fixed: $\tau_l = 7.5$ (baseline estimate) and $\xi_n = 0.01$ (solved endogenously). Model moments are computed from a simulated panel of 1,000 firms over 20,000 periods (500 burn-in), comparable to the firm-level panel data. Solved $c_e = 105.32$ at $p = 1$.

more and once the firm has furloughed, the probability of furloughing again is substantially higher than for firms that have not furloughed. Moreover, survey evidence supports an information-cost because firms report the legislation hard to navigate.

These patterns are jointly consistent with firms behaving as if they face a fixed cost the first time they furlough. A learning cost that requires the time, attention, and expertise to understand the legislation and set up furloughs is supported by the survey evidence. The model formalizes this microfoundation and disciplines its parameters with the empirical patterns above. The framework is designed to study questions such as whether the welfare improves if more firms are encouraged to take up furloughs. Two forces work in different directions - consumption and disutility of work.

The welfare effects of removing the learning channel are modest, at most just 0.01%. The disutility comes from labor-disutility, but consumption stays unchanged. The model confirms that furloughs are used to overcome expensive layoffs, as once the layoff costs are removed, furloughs are no longer used, and when the fixed cost of layoffs is removed, the use of furloughs doubles.

A natural extension of the model is to extend it to business cycle rather than focusing on the stationary equilibrium. There are several ways to extend the issues raised in this paper. One is to consider ex-ante heterogeneous firms in the spirit of Sterk et al. (2021). This would allow the welfare effects to vary over the business cycle. Moreover, this setup would allow one to study the importance of information frictions during recessions.

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A Appendix

A.1 Empirical

Table 12: Furlough rate conditional on Furloughing

	(1)	(2)	(3)	(4)
Constant	8.041***	8.041***	8.041***	8.008***
	(0.1094)	(0.0862)	(0.0848)	(0.0745)
Observations	310,548	310,548	310,546	283,834
R-squared	-0.000	0.021	0.029	0.465
Year FE		✓		
Industry FE			✓	
Firm FE				✓

Industry at 3-digit level. Standard errors are clustered at firm level. *p<0.10, **p<0.05, ***p<0.01.

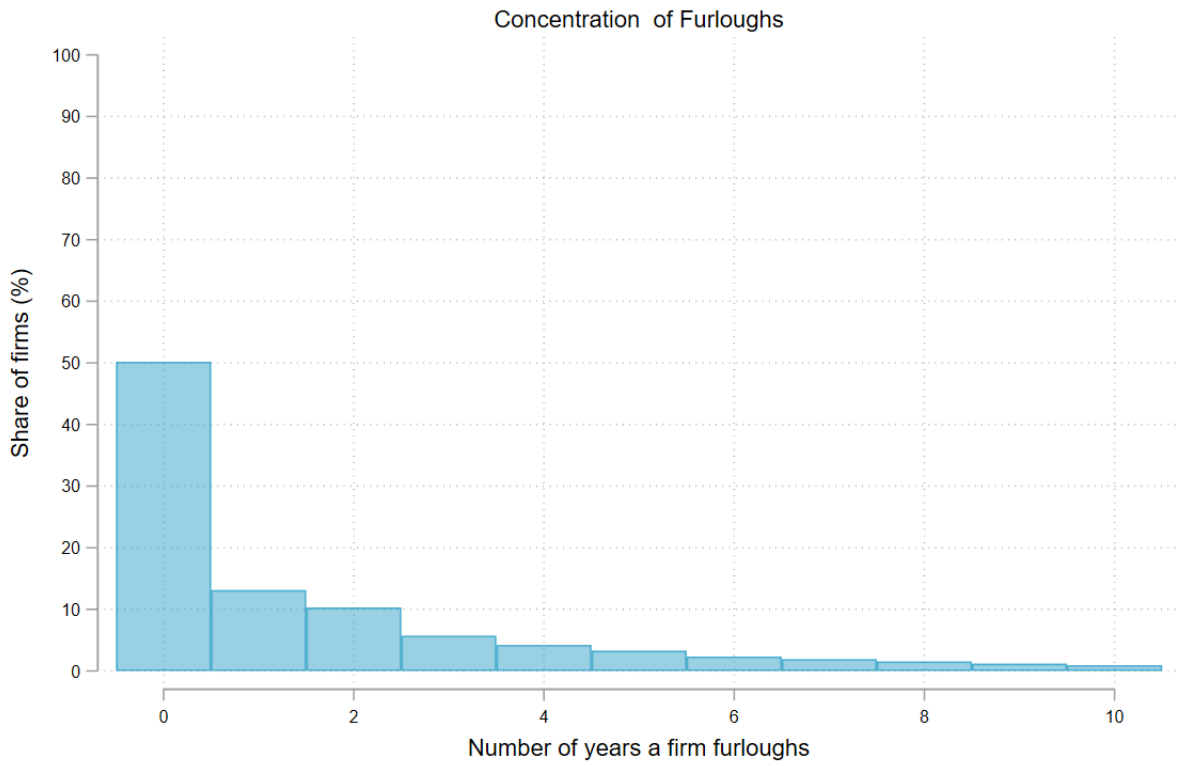


Figure 2: Fraction of firms that are in the sample for at least five years that furlough over the years 1994-2021.

Table 13: Furlough probability by size and productivity

Furlough rate	(1)	(2)	(3)	(4)
Employees	0.084***	0.084***	0.081***	0.090***
	(0.0016)	(0.0016)	(0.0019)	(0.0024)
Labor productivity	-0.040***	-0.036***	-0.043***	-0.043***
	(0.0023)	(0.0018)	(0.0016)	(0.0019)
Constant	0.053***	0.052***	0.058***	0.044***
	(0.0026)	(0.0025)	(0.0029)	(0.0033)
Observations	1,885,924	1,885,924	1,885,924	1,850,793
R^2	0.075	0.089	0.161	0.442
Year FE		✓		
Industry FE			✓	
Firm FE				✓

Industry at 3-digit level. Standard errors are clustered at firm level. *p<0.10, **p<0.05, ***p<0.01.

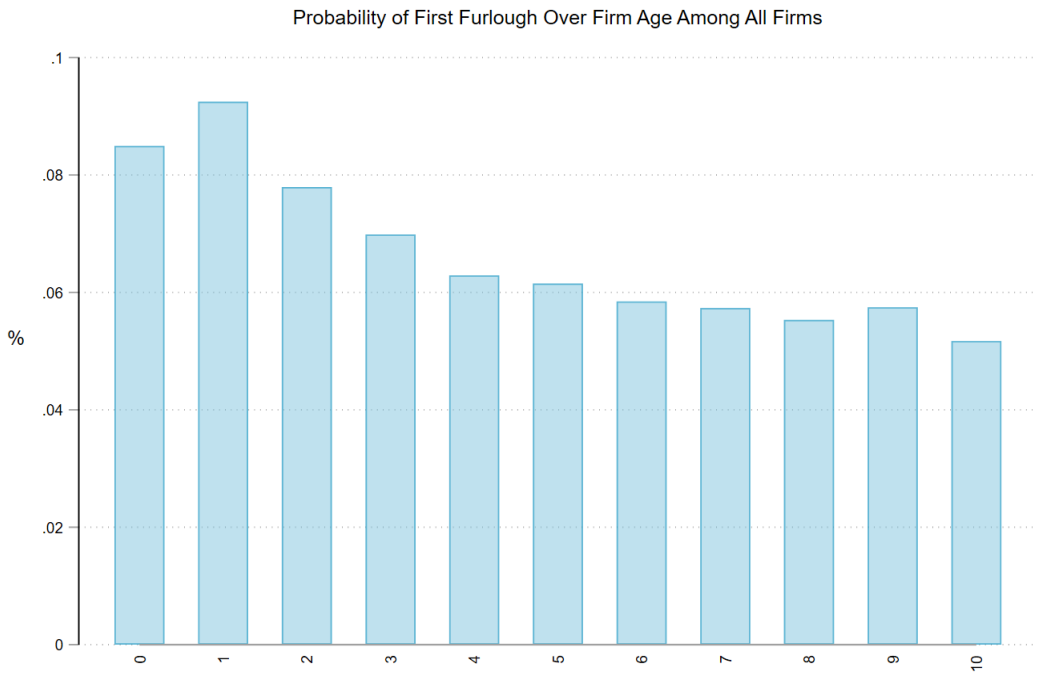


Figure 3: Hazard rate for first furlough over the years 1994-2021

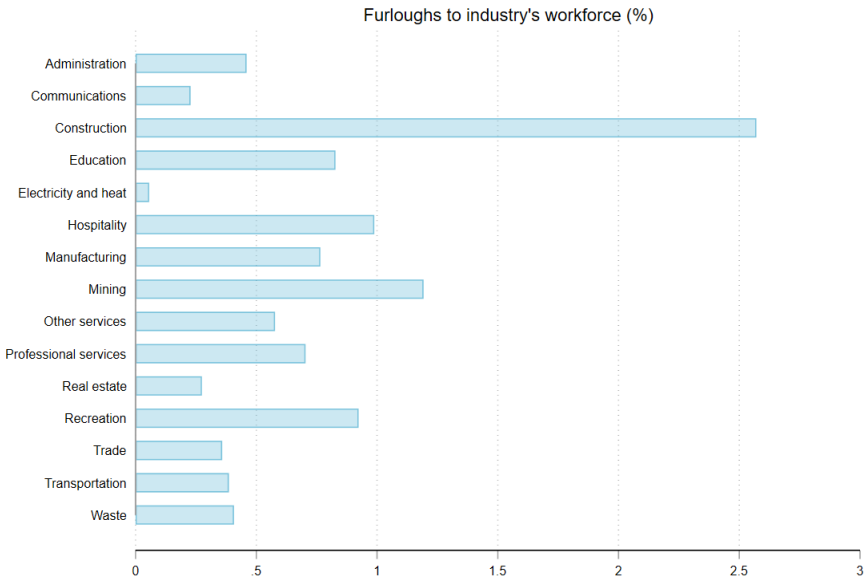


Figure 4: Fraction of furloughed workforce (in full-time equivalent) in each 1-digit industry, over the years 1994-2021

Table 14: Layoff probability

	(1)	(2)	(3)	(4)
Layoff Probability	0.134***	0.134***	0.134***	0.137***
	(0.0020)	(0.0012)	(0.0018)	(0.0019)
Observations	1,885,924	1,885,924	1,885,924	1,850,793
R-squared	0.000	0.025	0.015	0.172
Year FE		✓		
Industry FE			✓	
Firm FE				✓

Results from a linear probability model. Industry at 3-digit level. Standard errors are clustered at firm level. *p<0.10, **p<0.05, ***p<0.01.

Table 15: Layoff rate

	(1)	(2)	(3)	(4)
Layoff rate	3.693***	3.693***	3.693***	3.752***
	(0.0721)	(0.0403)	(0.0651)	(0.0674)
Observations	1,885,924	1,885,924	1,885,924	1,850,793
R-squared	-0.000	0.027	0.007	0.138
Year FE		✓		
Industry FE			✓	
Firm FE				✓

Industry at 3-digit level. Standard errors are clustered at firm level. *p<0.10, **p<0.05, ***p<0.01.

A.2 Layoffs and Fixed Effects

Contemporaneous revenue shocks. Table 16 regresses the furlough rate on the contemporaneous first difference of log revenue, controlling for size and productivity. The coefficient is large, negative, and stable across specifications: with firm fixed effects (column 4), a 10% worse shock is associated with a 0.12-0.13 percentage-point increase in the furlough share, roughly 9-10% of the unconditional mean.

A.3 Asymmetry in the shock response

The linear specification masks an asymmetry that is central to the story. Table 17 adds a squared term in $\Delta \log$ Revenue and yields a positive coefficient, which mechanically describes a U-shape but in fact reflects a kink at zero: the response to negative shocks is much stronger than to positive shocks. Splitting the sample by the sign of the revenue change (Table 18) makes this explicit. On the negative-shock subsample, the slope with firm fixed effects is -2.262, meaning that a 10% worsening of revenue is associated with

Table 16: Furlough rate and revenue shocks

Furlough rate	(1)	(2)	(3)	(4)
Δ log revenue	-1.235***	-1.174***	-1.278***	-1.349***
	(0.0593)	(0.0489)	(0.0576)	(0.0602)
Size	0.112***	0.098***	0.073***	0.155***
	(0.0094)	(0.0087)	(0.0111)	(0.0266)
Productivity	-0.500***	-0.431***	-0.496***	-0.429***
	(0.0408)	(0.0310)	(0.0296)	(0.0353)
Constant	1.376***	1.377***	1.437***	1.287***
	(0.0512)	(0.0462)	(0.0298)	(0.0485)
Observations	1,585,073	1,585,073	1,585,072	1,558,288
R^2	0.014	0.029	0.068	0.389
Year FE		✓		
Industry FE			✓	
Firm FE				✓

Industry at 3-digit level. Standard errors are clustered at firm level. *p<0.10, **p<0.05, ***p<0.01.

a 0.23-0.28 percentage-point increase in the furlough share. On the positive-shock subsample the slope is small in absolute value (-0.265 with firm fixed effects). The positive correlation with positive shocks on population level could result from the firm undergoing multiple shocks in a given year, as the data is annual.

A.4 Remove Firms that Furlough more than 80% of Time

A.5 Furlough Probability Persistence

A.6 Counterfactual Extra

To assess whether the learning channel explains the persistence of furloughs, I set the furlough setup cost τ_f to zero, which removes the channel. Column (2) of Table 37 reports the results. The share of firms furloughing in a given year increases from 18 to 46 percent, and nearly all firms (96 percent) furlough at some point. Most importantly, the persistence disappears. The persistence gap falls from 59 percentage points in the baseline to 1 percentage point, which is within simulation noise.

I also evaluate the learning channel at the two corner values of the forgetting probability δ_f. When $\delta_f = 0$ (column (3)), a firm pays the setup cost at most once over its lifetime. This lowers the expected cost of furloughing. As expected, the share of firms furloughing in a given year doubles, and the share that ever

Table 17: Furlough rate and revenue shocks, quadratic specification

Furlough rate	(1)	(2)	(3)	(4)
Δ log revenue	-1.302***	-1.241***	-1.327***	-1.394***
	(0.0606)	(0.0506)	(0.0591)	(0.0617)
(Δ log revenue)^2	0.429***	0.428***	0.319***	0.269***
	(0.0257)	(0.0243)	(0.0201)	(0.0200)
Size	0.118***	0.105***	0.079***	0.192***
	(0.0096)	(0.0088)	(0.0113)	(0.0279)
Productivity	-0.455***	-0.384***	-0.461***	-0.389***
	(0.0399)	(0.0303)	(0.0289)	(0.0346)
Constant	1.296***	1.296***	1.375***	1.182***
	(0.0481)	(0.0434)	(0.0289)	(0.0497)
Observations	1,585,073	1,585,073	1,585,072	1,558,288
R^2	0.016	0.032	0.069	0.390
Year FE		✓		
Industry FE			✓	
Firm FE				✓

Industry at 3-digit level. Standard errors are clustered at firm level. *p<0.10, **p<0.05, ***p<0.01.

Table 18: Furlough rate and asymmetric revenue shocks

	Negative shocks				Positive shocks			
Furlough rate	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Δ log revenue	-2.820***	-2.748***	-2.252***	-2.262***	0.547***	0.545***	0.144***	-0.265***
	(0.1288)	(0.1121)	(0.1164)	(0.1279)	(0.0382)	(0.0377)	(0.0199)	(0.0232)
Size	0.305***	0.267***	0.236***	0.455***	0.024***	0.021***	0.001	0.058***
	(0.0177)	(0.0146)	(0.0210)	(0.0438)	(0.0057)	(0.0058)	(0.0062)	(0.0166)
Productivity	-0.396***	-0.284***	-0.470***	-0.479***	-0.310***	-0.273***	-0.314***	-0.259***
	(0.0510)	(0.0367)	(0.0370)	(0.0427)	(0.0295)	(0.0243)	(0.0196)	(0.0260)
Constant	0.884***	0.935***	1.120***	0.796***	0.922***	0.915***	1.055***	1.025***
	(0.0429)	(0.0425)	(0.0340)	(0.0741)	(0.0363)	(0.0329)	(0.0216)	(0.0390)
Observations	712,445	712,445	712,444	674,345	872,628	872,628	872,628	837,407
R^2	0.026	0.049	0.077	0.453	0.003	0.011	0.056	0.431
Year FE		✓				✓		
Industry FE			✓				✓	
Firm FE				✓				✓

Sample split by sign of Δ log revenue. Columns (1)–(4) restrict to firm-years with Δ log revenue < 0; columns (5)–(8) to Δ log revenue > 0. Industry at 3-digit level. Standard errors are clustered at firm level. *p<0.10, **p<0.05, ***p<0.01.

Table 19: Furlough probability with liquidity and leverage measures

	Furlough probability					
	(1)	(2)	(3)	(4)	(5)	(6)
Liquidity	0.000					
	(0.0000)					
Cashflow		-0.000				
		(0.0000)				
Equity ratio			0.000			
			(0.0000)			
Short-term liabilities				-0.005***		
				(0.0007)		
Inventories					0.001	
					(0.0008)	
Interest rate						0.000**
						(0.0001)
Size	0.090***	0.090***	0.090***	0.095***	0.087***	0.092***
	(0.0024)	(0.0024)	(0.0024)	(0.0027)	(0.0031)	(0.0038)
Productivity	-0.043***	-0.043***	-0.043***	-0.044***	-0.052***	-0.053***
	(0.0019)	(0.0019)	(0.0019)	(0.0020)	(0.0028)	(0.0032)
Constant	0.044***	0.044***	0.044***	0.091***	0.052***	0.048***
	(0.0033)	(0.0033)	(0.0033)	(0.0062)	(0.0063)	(0.0053)
Observations	1,843,766	1,850,776	1,850,793	1,794,551	1,081,414	1,115,162
R^2	0.442	0.442	0.442	0.445	0.452	0.469

Extensive margin: dependent variable is an indicator for furloughing in a given period. Regression includes firm fixed effects. Standard errors are clustered at firm level. *p<0.10, **p<0.05, ***p<0.01.

furlough increases from 33 to 54 percent. The persistence gap falls from 59 to 48 percentage points but remains large. Even when the setup cost is paid only once, learning generates persistence. When $\delta_f = 1$ (column (4)), a firm loses its experience whenever it does not furlough. To avoid paying the setup cost again, a firm that has furloughed has an incentive to keep furloughing. The persistence gap widens to 68 percentage points: firms sort more strongly into those that furlough repeatedly and those that never do. The share of firms furloughing in a given year and the share that ever furlough both fall below their baseline values.

The speed of learning does not change the firm size distribution, and the welfare effects are small. Relative to the baseline, welfare changes by between -0.010 and 0.062 percent.

Table 20: Furlough rate with various liquidity and leverage measures, no fixed effects

	Furlough rate					
	(1)	(2)	(3)	(4)	(5)	(6)
Liquidity	-0.000					
	(0.0000)					
Cashflow		-0.000*				
		(0.0000)				
Equity ratio			-0.000			
			(0.0000)			
Short-term debt				-0.025***		
				(0.0071)		
Current assets					-0.058***	
					(0.0127)	
Interest rate						-0.000***
						(0.0001)
Constant	1.325***	1.324***	1.324***	1.590***	2.023***	1.443***
	(0.0451)	(0.0450)	(0.0450)	(0.1116)	(0.1688)	(0.0637)
Observations	1,878,848	1,885,907	1,885,924	1,828,709	1,111,498	1,142,366
R^2	0.000	0.000	0.000	0.000	0.001	0.000

No fixed effects. Standard errors are clustered at firm level. *p<0.10, **p<0.05, ***p<0.01.

Table 21: Furlough probability in firms that furlough less than 80% of time

Furlough probability	(1)	(2)	(3)	(4)
Constant	0.133***	0.133***	0.133***	0.135***
	(0.0030)	(0.0027)	(0.0019)	(0.0019)
Observations	1,809,705	1,809,705	1,809,705	1,777,532
R^2	0	0.0176	0.0607	0.3194
Year FE		✓		
Industry FE			✓	
Firm FE				✓

Extensive margin: dependent variable is an indicator for furloughing in a given period. Sample consists of firms that furlough less than 80% of the time. Industry at 3-digit level. Standard errors are clustered at firm level. *p<0.10, **p<0.05, ***p<0.01.

Table 22: Furlough rate in firms that furlough less than 80% of time

Furlough rate	(1)	(2)	(3)	(4)
Constant	1.005***	1.005***	1.005***	1.023***
	(0.0343)	(0.0301)	(0.0271)	(0.0262)
Observations	1,809,705	1,809,705	1,809,705	1,777,532
R^2	0	0.0172	0.0271	0.2507
Year FE		✓		
Industry FE			✓	
Firm FE				✓

Intensive margin: dependent variable is the furlough rate. Sample consists of firms that furlough less than 80% of the time. Industry at 3-digit level. Standard errors are clustered at firm level. *p<0.10, **p<0.05, ***p<0.01.

Table 23: Furlough rate in firms that furlough less than 80% of time

Furlough rate	(1)	(2)	(3)	(4)
Employees	0.071***	0.068***	0.065***	0.253***
	(0.0082)	(0.0083)	(0.0098)	(0.0226)
Labor productivity	-0.420***	-0.367***	-0.452***	-0.577***
	(0.0342)	(0.0259)	(0.0259)	(0.0363)
Constant	1.019***	1.008***	1.036***	0.817***
	(0.0355)	(0.0316)	(0.0241)	(0.0394)
Observations	1,809,705	1,809,705	1,809,705	1,777,532
R^2	0.003	0.020	0.031	0.254
Year FE		✓		
Industry FE			✓	
Firm FE				✓

Intensive margin: dependent variable is the furlough rate. Sample consists of firms that furlough less than 80% of the time. Industry at 3-digit level. Standard errors are clustered at firm level. *p<0.10, **p<0.05, ***p<0.01.

Table 24: Furlough probability in firms that furlough less than 80% of time

Furlough probability	(1)	(2)	(3)	(4)
Employees	0.066***	0.065***	0.065***	0.088***
	(0.0013)	(0.0013)	(0.0015)	(0.0024)
Labor productivity	-0.034***	-0.030***	-0.038***	-0.043***
	(0.0021)	(0.0016)	(0.0014)	(0.0020)
Constant	0.049***	0.048***	0.050***	0.020***
	(0.0022)	(0.0022)	(0.0021)	(0.0032)
Observations	1,809,705	1,809,705	1,809,705	1,777,532
R^2	0.052	0.069	0.108	0.335
Year FE		✓		
Industry FE			✓	
Firm FE				✓

Extensive margin: dependent variable is an indicator for furloughing in a given period. Sample consists of firms that furlough less than 80% of the time. Industry at 3-digit level. Standard errors are clustered at firm level. *p<0.10, **p<0.05, ***p<0.01.

Table 25: Furlough probability and revenue shocks in firms that furlough less than 80% of time

Furlough probability	(1)	(2)	(3)	(4)
Δ log revenue	-0.057***	-0.052***	-0.061***	-0.071***
	(0.0030)	(0.0023)	(0.0027)	(0.0031)
Size	0.065***	0.064***	0.064***	0.092***
	(0.0013)	(0.0012)	(0.0015)	(0.0026)
Productivity	-0.039***	-0.035***	-0.041***	-0.036***
	(0.0025)	(0.0020)	(0.0017)	(0.0020)
Constant	0.057***	0.057***	0.058***	0.015***
	(0.0027)	(0.0025)	(0.0023)	(0.0039)
Observations	1,520,620	1,520,620	1,520,619	1,495,452
R^2	0.050	0.070	0.113	0.357
Year FE		✓		
Industry FE			✓	
Firm FE				✓

Extensive margin: dependent variable is an indicator for furloughing in a given period. Sample consists of firms that furlough less than 80% of the time. Industry at 3-digit level. Standard errors are clustered at firm level. *p<0.10, **p<0.05, ***p<0.01.

Table 26: Furlough probability and persistent revenue volatility in firms that furlough less than 80% of time

Furlough probability	(1)	(2)	(3)	(4)
SD of Δ log revenue	0.126***	0.126***	0.071***	0.037***
	(0.0051)	(0.0049)	(0.0030)	(0.0028)
Size	0.064***	0.063***	0.063***	0.083***
	(0.0013)	(0.0012)	(0.0015)	(0.0025)
Productivity	-0.050***	-0.043***	-0.055***	-0.061***
	(0.0029)	(0.0022)	(0.0020)	(0.0028)
Constant	0.032***	0.032***	0.049***	0.025***
	(0.0025)	(0.0023)	(0.0024)	(0.0039)
Observations	1,312,370	1,312,370	1,312,370	1,293,378
R^2	0.054	0.076	0.114	0.363
Year FE		✓		
Industry FE			✓	
Firm FE				✓

Extensive margin: dependent variable is an indicator for furloughing in a given period. Sample consists of firms that furlough less than 80% of the time. Volatility measured as 3-year rolling standard deviation of Δ log revenue. Industry at 3-digit level. Standard errors are clustered at firm level. *p<0.10, **p<0.05, ***p<0.01.

Table 27: Furlough rate and revenue shocks in firms that furlough less than 80% of time

Furlough rate	(1)	(2)	(3)	(4)
Δ log revenue	-1.050***	-0.992***	-1.082***	-1.191***
	(0.0561)	(0.0458)	(0.0547)	(0.0585)
Size	0.047***	0.032***	0.039***	0.253***
	(0.0087)	(0.0072)	(0.0101)	(0.0272)
Productivity	-0.422***	-0.357***	-0.431***	-0.414***
	(0.0386)	(0.0292)	(0.0281)	(0.0352)
Constant	1.140***	1.144***	1.155***	0.827***
	(0.0419)	(0.0373)	(0.0271)	(0.0479)
Observations	1,520,620	1,520,620	1,520,619	1,495,452
R^2	0.012	0.032	0.043	0.291
Year FE		✓		
Industry FE			✓	
Firm FE				✓

Intensive margin: dependent variable is the furlough rate. Sample consists of firms that furlough less than 80% of the time. Industry at 3-digit level. Standard errors are clustered at firm level. *p<0.10, **p<0.05, ***p<0.01.

Table 28: Furlough rate and persistent revenue volatility in firms that furlough less than 80% of time

Furlough rate	(1)	(2)	(3)	(4)
SD of Δ log revenue	1.475***	1.491***	1.031***	0.679***
	(0.0716)	(0.0661)	(0.0574)	(0.0537)
Size	0.013*	-0.001	-0.008	0.095***
	(0.0076)	(0.0067)	(0.0086)	(0.0265)
Productivity	-0.603***	-0.507***	-0.648***	-0.809***
	(0.0487)	(0.0367)	(0.0369)	(0.0504)
Constant	0.899***	0.893***	1.048***	1.005***
	(0.0377)	(0.0334)	(0.0228)	(0.0505)
Observations	1,312,370	1,312,370	1,312,370	1,293,378
R^2	0.014	0.036	0.041	0.296
Year FE		✓		
Industry FE			✓	
Firm FE				✓

Intensive margin: dependent variable is the furlough rate. Sample consists of firms that furlough less than 80% of the time. Volatility measured as 3-year rolling standard deviation of Δ log revenue. Industry at 3-digit level. Standard errors are clustered at firm level. *p<0.10, **p<0.05, ***p<0.01.

Table 29: Furlough rate with various liquidity and leverage measures in firms that furlough less than 80% of time

	Furlough rate					
	(1)	(2)	(3)	(4)	(5)	(6)
Liquidity	0.000					
	(0.0000)					
Cashflow		-0.000				
		(0.0002)				
Equity ratio			-0.000			
			(0.0000)			
Short-term liabilities				-0.131***		
				(0.0115)		
Inventories					-0.016	
					(0.0107)	
Interest rate						0.000**
						(0.0002)
Size	0.253***	0.253***	0.253***	0.361***	0.209***	0.257***
	(0.0226)	(0.0226)	(0.0226)	(0.0301)	(0.0394)	(0.0427)
Productivity	-0.579***	-0.577***	-0.577***	-0.583***	-0.668***	-0.773***
	(0.0363)	(0.0363)	(0.0363)	(0.0374)	(0.0520)	(0.0592)
Constant	0.819***	0.817***	0.817***	2.160***	1.101***	0.950***
	(0.0395)	(0.0394)	(0.0394)	(0.1069)	(0.0872)	(0.0701)
Observations	1,770,718	1,777,515	1,777,532	1,724,164	1,031,679	1,072,110
R^2	0.254	0.254	0.254	0.259	0.264	0.301

Intensive margin: dependent variable is the furlough rate. Sample consists of firms that furlough less than 80% of the time. Regression includes firm fixed effects. Standard errors are clustered at firm level. *p<0.10, **p<0.05, ***p<0.01.

Table 30: Furlough probability with various liquidity and leverage measures in firms that furlough less than 80% of time

	Furlough probability					
	(1)	(2)	(3)	(4)	(5)	(6)
Liquidity	0.000					
	(0.0000)					
Cashflow		-0.000				
		(0.0000)				
Equity ratio			0.000			
			(0.0000)			
Short-term liabilities				-0.004***		
				(0.0007)		
Inventories					0.001	
					(0.0008)	
Interest rate						0.000**
						(0.0001)
Size	0.088***	0.088***	0.088***	0.093***	0.086***	0.091***
	(0.0024)	(0.0024)	(0.0024)	(0.0028)	(0.0032)	(0.0039)
Productivity	-0.043***	-0.043***	-0.043***	-0.044***	-0.053***	-0.054***
	(0.0020)	(0.0020)	(0.0020)	(0.0020)	(0.0028)	(0.0032)
Constant	0.020***	0.020***	0.020***	0.063***	0.021**	0.024***
	(0.0032)	(0.0032)	(0.0032)	(0.0062)	(0.0066)	(0.0052)
Observations	1,770,718	1,777,515	1,777,532	1,724,164	1,031,679	1,072,110
R^2	0.335	0.335	0.335	0.338	0.345	0.375

Extensive margin: dependent variable is an indicator for furloughing in a given period. Sample consists of firms that furlough less than 80% of the time. Regression includes firm fixed effects. Standard errors are clustered at firm level. *p<0.10, **p<0.05, ***p<0.01.

Table 31: Furlough probability given that the firm has not furloughed in the first five years: additional controls

Furlough probability	(1)	(2)	(3)
Employees	0.045***	0.045***	0.048***
	(0.0006)	(0.0006)	(0.0006)
Labor productivity	-0.017***	-0.020***	-0.021***
	(0.0011)	(0.0011)	(0.0012)
Constant	0.037***	0.038***	0.034***
	(0.0010)	(0.0010)	(0.0010)
Observations	248,493	248,493	248,492
R^2	0.028	0.048	0.083
Year FE		✓	
Industry FE			✓

Dependent variable is the furlough probability over the next five years, given that the firm has not furloughed in its first five years of life. Early years refers to the first five years of the firm's lifetime. Industry at 3-digit level. Standard errors are clustered at firm level. *p<0.10, **p<0.05, ***p<0.01.

Table 32: Furlough probability given that the firm has furloughed in the first five years: additional controls

Furlough probability	(1)	(2)	(3)
Employees	0.075***	0.075***	0.085***
	(0.0015)	(0.0015)	(0.0016)
Labor productivity	-0.059***	-0.053***	-0.071***
	(0.0049)	(0.0049)	(0.0048)
Constant	0.283***	0.281***	0.266***
	(0.0040)	(0.0040)	(0.0040)
Observations	62,678	62,678	62,678
R^2	0.041	0.053	0.134
Year FE		✓	
Industry FE			✓

Dependent variable is the furlough probability over the next five years, given that the firm has furloughed in its first five years of life. Early years refers to the first five years of the firm's lifetime. Industry at 3-digit level. Standard errors are clustered at firm level. *p<0.10, **p<0.05, ***p<0.01.

Table 33: Furlough probability given that the firm has not furloughed in the first five years

Furlough probability	(1)	(2)	(3)
Constant	0.140***	0.140***	0.140***
	(0.0003)	(0.0003)	(0.0003)
Observations	1,103,960	1,103,960	1,103,960
R^2	-0.000	0.017	0.086
Year FE		✓	
Industry FE			✓

Dependent variable is an indicator for furloughing in a given period. Early years refers to the first five years of the firm's lifetime. Industry at 3-digit level. Standard errors are clustered at firm level. *p<0.10, **p<0.05, ***p<0.01.

Table 34: Furlough probability given that the firm has not furloughed in the first five years

Furlough probability	(1)	(2)	(3)
Employees	0.071***	0.070***	0.070***
	(0.0003)	(0.0003)	(0.0003)
Labor productivity	-0.035***	-0.034***	-0.040***
	(0.0006)	(0.0006)	(0.0006)
Constant	0.045***	0.045***	0.048***
	(0.0006)	(0.0006)	(0.0006)
Observations	1,103,960	1,103,960	1,103,960
R^2	0.060	0.075	0.139
Year FE		✓	
Industry FE			✓

Dependent variable is an indicator for furloughing in a given period. Early years refers to the first five years of the firm's lifetime. Industry at 3-digit level. Standard errors are clustered at firm level. *p<0.10, **p<0.05, ***p<0.01.

Table 35: Furlough probability given that the firm has furloughed in the first five years

Furlough probability	(1)	(2)	(3)
Constant	0.408***	0.408***	0.408***
	(0.0012)	(0.0012)	(0.0012)
Observations	156,710	156,710	156,710
R^2	0.000	0.016	0.094
Year FE		✓	
Industry FE			✓

Dependent variable is an indicator for furloughing in a given period. Early years refers to the first five years of the firm's lifetime. Industry at 3-digit level. Standard errors are clustered at firm level. *p<0.10, **p<0.05, ***p<0.01.

Table 36: Furlough probability given that the firm has furloughed in the first five years

Furlough probability	(1)	(2)	(3)
Employees	0.083***	0.083***	0.093***
	(0.0009)	(0.0009)	(0.0009)
Labor productivity	-0.069***	-0.064***	-0.076***
	(0.0029)	(0.0029)	(0.0028)
Constant	0.250***	0.248***	0.231***
	(0.0024)	(0.0024)	(0.0024)
Observations	156,710	156,710	156,710
R^2	0.055	0.070	0.153
Year FE		✓	
Industry FE			✓

Dependent variable is an indicator for furloughing in a given period. Early years refers to the first five years of the firm's lifetime. Industry at 3-digit level. Standard errors are clustered at firm level. *p<0.10, **p<0.05, ***p<0.01.

Table 37: Counterfactuals: importance of the learning channel

Moment	Baseline	No learning	Persistent learning	Forget if not used
	(1)	(2)	(3)	(4)
		$\tau_f = 0$	$\delta_f = 0$	$\delta_f = 1$
Share of firm-years furloughing	0.18	0.46	0.37	0.11
Share ever furloughing	0.33	0.96	0.54	0.25
Furlough intensity	0.04	0.04	0.04	0.04
P(furlough | early furlough)	0.64	0.43	0.60	0.71
P(furlough | no early furlough)	0.05	0.42	0.12	0.03
Persistence gap	0.59	0.01	0.48	0.68
Share of gap removed	–	98%	18%	−15%
Mean employment	57.81	57.73	57.96	57.69
Size 1–19	0.59	0.59	0.59	0.59
Size 20–99	0.29	0.30	0.30	0.29
Size 100–499	0.10	0.10	0.10	0.10
Size 500+	0.02	0.02	0.02	0.02
GE price p^*	1.0000	0.9995	0.9999	1.0000
Δ Welfare (%)	–	0.062	0.015	−0.010
Consumption (%)	–	0.051	0.009	−0.003
Labor disutility (%)	–	0.011	0.006	−0.007

Notes: Firm-year furloughs, intensity, size shares and welfare are computed from the stationary distribution; mean employment, ever-furlough shares and the early/no-early furlough probabilities (years 6–10 conditional on furloughing or not in years 1–5) are simulated. Persistence gap = P(furlough | early) − P(furlough | no early); share of gap removed = $(\text{gap}_0 - \text{gap}_k)/\text{gap}_0$.